



Nikita Chaurasia

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[Leetcode](#) | [GFG](#) | [LinkedIn](#) | [GitHub](#)

EDUCATION

B.Tech - ECAM (ECE + AI/ML)	2023-2027	Netaji Subhas University of Technology	7.03
CBSE - Class XII	2023	Rani Jhansi Sarvodaya Kanya Vidhyalaya, New Delhi	83%
CBSE - Class X	2021	Rani Jhansi Sarvodaya Kanya Vidhyalaya, New Delhi	95.5%

ACADEMIC PROJECTS

SENTINEL — AI Prompt Security Gateway

June 2026

Live Demo - nikitayk.github.io/SENTINEL

GitHub - <https://github.com/nikitayk/SENTINEL>

- Built a **Manifest V3** Chrome extension that intercepts prompts on ChatGPT, Claude, and Gemini, scanning for **API keys, credentials, and PII** before submission with **zero network calls** in the detection path.
- Implemented **5 algorithms from scratch** - Aho-Corasick, Shannon entropy, Trie, Max-Heap, and Merge-Intervals - detecting **24 secret patterns** plus base64-encoded payloads via a decode-and-rescan layer.
- Shipped a React + Chart.js dashboard backed by a FastAPI + JWT service, **3 tunable sensitivity levels**, and **104 tests** covering ReDoS safety and a false-positive corpus.

ADPULSE — Real Time Demand Side Platform

June 2026

Deployed Link - adpulse-457r.onrender.com

GitHub - <https://github.com/nikitayk/ADPULSE>

- Built a full-stack **Demand-Side Platform (DSP)** that participates in real-time ad auctions, prices each bid in-memory at **~5 ms**, and sustains a 7 bids/sec live feed.
- Trained **2 LightGBM models (CTR + CVR)** on IPinYou competition logs with 12 engineered features, bid pricing scales up to **11x base price** with a budget pacer and a hard cap of 300.
- Engineered a generator-based pipeline that streams **multi-GB logs in O(1) memory**, a Flask-SocketIO + **Three.js 3D dashboard** renders live KPIs, validated by **46 tests** on a Python 3.9/3.11 CI matrix.

ACHIEVEMENTS AND AWARDS

- Solved **300+** Data Structures and Algorithms problems on LeetCode and GeeksforGeeks.
- Ranked in the **top 4% out of 2,500+ teams** at DTU Execute 4.0 Hackathon.
- **Top 20 at Hackmania** Microsoft Azure & The Elites at Microsoft Gurugram.

TECHNICAL SKILLS

- **Programming:** Python, C++, SQL
- **CS Subjects:** Data Structures & Algorithms (DSA), Operating Systems, Database Management Systems (DBMS), Computer Networks, Object Oriented Programming (OOP)
- **Problem-Solving:** Algorithm Design, Complexity Analysis, Debugging & Optimization
- **Analytical Skills:** Logical Reasoning, Pattern Recognition, Data Interpretation

CERTIFICATIONS

- AI For Everyone – Deep Learning.AI
- Supervised Machine Learning (Regression and Classification) - Stanford Online
- Advanced Learning Algorithms – DeepLearning.AI